

DNA Repair Defects and Complementation Groups in Xeroderma Pigmentosum (XP)

Most XP cells exhibit a normal response to x-ray treatment, highlighting the specificity of the DNA repair defect to ultraviolet (UV) radiation-induced damage. The defective genes responsible for the seven nucleotide excision repair (NER)-deficient forms of XP, as well as the XP variant, have been cloned, and their functions are under active investigation (see Chap. 109).

Complementation Groups

Genetic heterogeneity among XP patients has been characterized through cell fusion experiments using cultured fibroblasts, which allowed the identification of distinct complementation groups (see Chap. 109). As of 2006, seven complementation groups—designated **XP-A** to **XP-G**—have been identified, each corresponding to a specific gene involved in NER (see Fig. 140-1). In addition to these groups, patients with clinical features of XP but intact NER are classified as **XP variants**. Cellular studies show these variant cells exhibit mild hypersensitivity to UV-induced inhibition of cell growth, an effect enhanced by caffeine. This defect is linked to mutations in **DNA polymerase eta (pol η)**, an error-prone polymerase responsible for translesion synthesis across UV-damaged DNA.

Prenatal Diagnosis

Prenatal diagnosis of XP is feasible through two primary methods: measurement of UV-induced unscheduled DNA synthesis in cultured amniotic fluid cells, and molecular genetic testing of trophoblast cells obtained early in pregnancy.

Drug and Chemical Hypersensitivity

Beyond UV radiation, XP cells show increased sensitivity to various DNA-damaging agents. These include: - **Psoralens** and **chlorpromazine** -

Chemotherapeutic agents such as cisplatin and carmustine - **Chemical carcinogens** like benzo[a]pyrene derivatives

The hypersensitivity likely arises because repair of the DNA damage induced by these agents involves pathways overlapping with the NER system, which is defective in XP.

Treatment and Management

Management of XP focuses on early diagnosis, rigorous UV protection, and prompt detection and treatment of skin neoplasms. Diagnosis is based on clinical features and confirmed by laboratory tests demonstrating cellular hypersensitivity to UV radiation and defective DNA repair. Molecular genetic testing for causative mutations is increasingly transitioning from research settings to clinical diagnostics.

Sun Protection

Patients must be educated to minimize UV exposure through comprehensive protective measures: - Wearing **protective clothing**, **UV-absorbing sunglasses**, and **long hairstyles** - Adopting lifestyle modifications to avoid sunlight - Daily use of broad-spectrum sunscreens with a minimum **SPF of 15**

Regular skin surveillance is critical: - Frequent self-examinations guided by trained family members - Periodic full-body skin examinations by a physician (every 3–6 months, depending on disease severity) - Documentation using **color photographs** of the entire skin surface, including close-ups of lesions with a ruler for size reference

Treatment of Premalignant and Malignant Lesions

- **Actinic keratoses** (premalignant lesions) can be treated with:
- Cryotherapy (liquid nitrogen)
- Topical **5-fluorouracil**
- Topical **imiquimod**

- **Photodynamic therapy (PDT)** using topical 5-aminolevulinic acid followed by blue light exposure is effective for actinic keratoses in the general population. However, **no safety or efficacy data exist for XP patients**, and caution is advised due to the potential for abnormal cellular responses to light-based therapies.
- For extensive skin damage, **therapeutic dermatome shaving** or **dermabrasion** may be used to remove damaged epidermal layers. This promotes repopulation of the skin by relatively UV-protected cells from hair follicles and skin glands.

Protection from Environmental Mutagens

Given the hypersensitivity of XP cells to chemical mutagens such as **benzo[a]pyrene** in cigarette smoke, patients should avoid exposure to tobacco and other environmental carcinogens. Clinical evidence supports this recommendation: one XP patient who smoked for over 10 years developed bronchogenic carcinoma at age 35, and another smoker developed lung cancer at age 48.